Software Requirements Specification

AI Chatbot for Patient Registration – PES Hospital

1. Introduction

This document describes the software requirements for an AI-powered chatbot built as an internship project for PES University Hospital. The goal of this document is to clearly explain the problem we are solving, what our chatbot will do, how users will interact with it, and how we plan to build it using the MERN stack.

2. Problem Statement

PES Hospital receives around 1,800 to 2,000 outpatients on a typical day. The current registration process works like this:

1. The patient walks in and goes to the OPD registration counter.
2. The counter staff collects their Aadhaar or government ID, name, age, and complaint.
3. Based on the complaint, the staff decides which OPD department to allocate the patient to.
4. The patient gets a slip and heads to that department.

This creates three real problems:

- Long queues at the counter, especially in the morning peak hours.
- Language barrier — patients speak Kannada, Telugu, Tamil, Hindi, and other languages. Counter staff cannot always understand everyone, which leads to errors in department allocation.
- Manual data entry is slow and can result in duplicate records or misallocations.

We are not trying to replace the counter. The counter will still exist and handle difficult cases, elderly patients, and emergencies. Our chatbot simply reduces the load on the counter by letting patients who are comfortable with a phone or tablet register themselves.

3. Our Approach

We will build a web-based chatbot that patients can access by scanning a QR code at the hospital entrance or by using a kiosk tablet provided by the hospital. No app installation is needed.

The chatbot will:

- Ask the patient to choose their preferred language.
- Ask whether they are a new or returning patient.
- Collect basic details (name, age, ID, complaint) through a simple conversation.
- Suggest the correct OPD department based on their complaint.
- Generate a token number so they can go directly to that OPD floor.

On the counter side, staff will have a simple dashboard that shows all chatbot-registered patients in a queue. This reduces the need to manually collect and enter data for every patient.

4. Chatbot Flow

Below is the step-by-step flow of how the chatbot works:

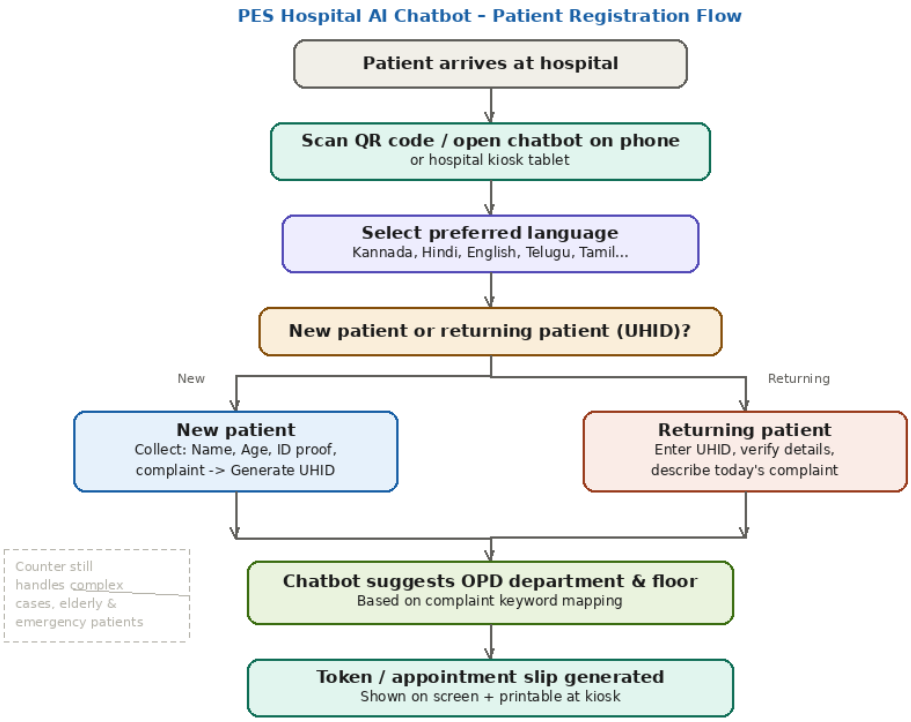

Figure 1: Patient Registration Flow through the Chatbot

Step-by-step explanation

5. Patient arrives at the hospital and scans the QR code on their phone, or uses the kiosk tablet at the entrance (if provided).
6. The chatbot opens in the browser — no login or account needed.
7. Patient selects their preferred language (Kannada, Hindi, English, Telugu, or Tamil).
8. The chatbot asks: "Are you visiting for the first time, or have you visited before?"
9. If new patient: chatbot collects name, age, gender, ID type, ID number, and complaint. A unique Patient ID (UHID) is generated and stored.
10. If returning patient: patient enters their UHID, confirms their details, and describes today's complaint.
11. The chatbot matches the complaint to a department using a keyword lookup and tells the patient which OPD floor and room to go to.
12. A token number is shown on screen. The patient proceeds to the OPD floor directly.

5. What the Chatbot Does

5.1 For the patient

- Greets the patient in their chosen language.
- Guides them through registration step by step — they just answer simple questions.
- Reads questions aloud using voice (text-to-speech) so low-literacy users can follow along.
- Accepts voice input — patients can speak their complaint instead of typing.
- Tells them exactly which department and floor to go to, with a token number.

5.2 For the counter staff

- A simple web dashboard shows all chatbot-registered patients for the day.
- Staff can see patient name, complaint, assigned department, and token number.
- Staff can still manually register patients who could not use the chatbot.

5.3 Department allocation logic

We will use a keyword-to-department mapping stored in our database. When a patient types or speaks their complaint, we check the keywords and assign the matching OPD department. Examples:

Complaint Keywords	OPD Department
Fever, cold, cough, body ache, weakness	**General Medicine**
Chest pain, palpitations, breathlessness	**Cardiology**
Eye pain, blurry vision, redness	**Ophthalmology**
Stomach pain, vomiting, diarrhoea, acidity	**Gastroenterology**
Bone pain, joint pain, fracture, back pain	**Orthopaedics**
Skin rash, itching, eczema, hair loss	**Dermatology**
Child fever, vaccination, child growth	**Paediatrics**
Toothache, gum pain, cavity	**Dental**
Anything not matched above	**General Medicine (default)**

6. Language and Voice Support

Since language barriers are the core problem, we are giving this feature priority. The chatbot will support:

- English, Kannada, Hindi, Telugu, and Tamil at minimum.
- All UI text and chatbot messages will change when a language is selected.
- Voice output (text-to-speech) will read each question aloud using the browser's built-in Speech Synthesis API — no paid API needed.
- Voice input (speech-to-text) using the browser's Web Speech API — patient can speak their complaint.

7. Technology Stack (MERN)

Our team has decided to use the MERN stack as it is what we are most comfortable with:

Layer	Technology	What it does in our project
Frontend	React.js + Tailwind CSS	Chatbot UI, language switching, patient-facing screens
Backend	Node.js + Express.js	API that handles registration logic and OPD mapping
Database	MongoDB (Atlas free tier)	Stores patient records, UHIDs, and department mapping
Voice	Web Speech API (browser built-in)	Free — no API key needed for voice input/output
Multilingual	i18next library	Manages translation JSON files for each language
Hosting	Vercel (frontend) + Render (backend)	Both have free tiers — good for a student project
QR Code	qrcode.js	Generates the QR code that launches the chatbot

8. What We Are Not Building

To keep the project realistic for an internship timeline, these are intentionally out of scope:

- We are not integrating with the hospital's existing software systems.
- We are not building an appointment booking or doctor scheduling feature.
- We are not making a mobile app — it is a web application only.
- We are not doing medical diagnosis — the chatbot only suggests a department, not a treatment.
- We are not building a payment or billing system.